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Experiment 10

Aim

Exploratory Data Analysis of a given text (Word Cloud).

Theory

Exploratory Data Analysis (EDA) is the process of analyzing and visualizing data to understand its main characteristics, patterns, and underlying structure before applying further NLP techniques. For text data, EDA typically involves:

1. **Basic statistics** – Number of documents, word counts, character counts, average document length.
2. **Category/class distribution** – Understanding how the text data is spread across different classes or labels.
3. **Word frequency analysis** – Identifying the most frequently occurring words after removing stopwords.
4. **Word Cloud visualization** – A visual representation of text data where the size of each word indicates its frequency; more frequent words appear larger and bolder.

A **Word Cloud** is one of the most popular EDA tools for text data because it gives an instant, intuitive sense of which words dominate a document or corpus, without having to read through all the text manually.

Application

- Quickly summarizing large text corpora (resumes, reviews, articles, tweets)
- Identifying dominant themes/keywords in a dataset
- Comparing vocabulary across different categories or classes
- Preliminary step before building NLP models (helps spot noisy or irrelevant frequent terms)

Advantages

- Simple, fast, and visually intuitive
- Requires no complex modeling — good first step in any text analysis pipeline
- Helps identify important keywords and irrelevant noise (e.g. common but meaningless words)

Limitations

- Does not capture context, grammar, or word relationships
- Can be misleading if stopwords/noise aren't properly removed
- Word size is based purely on frequency, not importance or sentiment

In [2]:
```
import pandas as pd
import nltk
import re
import matplotlib.pyplot as plt
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from wordcloud import WordCloud
from collections import Counter

nltk.download('punkt')
nltk.download('stopwords')
```

```
[nltk_data] Downloading package punkt to /home/computer/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data]     /home/computer/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
```

Out[2]:
```
True
```

In [4]:
```
df = pd.read_csv("UpdatedResumeDataSet.csv")
print("Shape :", df.shape)
df.head()
```

```
Shape : (962, 2)
```

Out[4]:

	Category	Resume
0	Data Science	Skills * Programming Languages: Python (pandas...
1	Data Science	Education Details \r\nMay 2013 to May 2017 B.E...
2	Data Science	Areas of Interest Deep Learning, Control Syste...
3	Data Science	Skills â¯¢ R â¯¢ Python â¯¢ SAP HANA â¯¢ Table...
4	Data Science	Education Details \r\n MCA YMCAUST, Faridab...

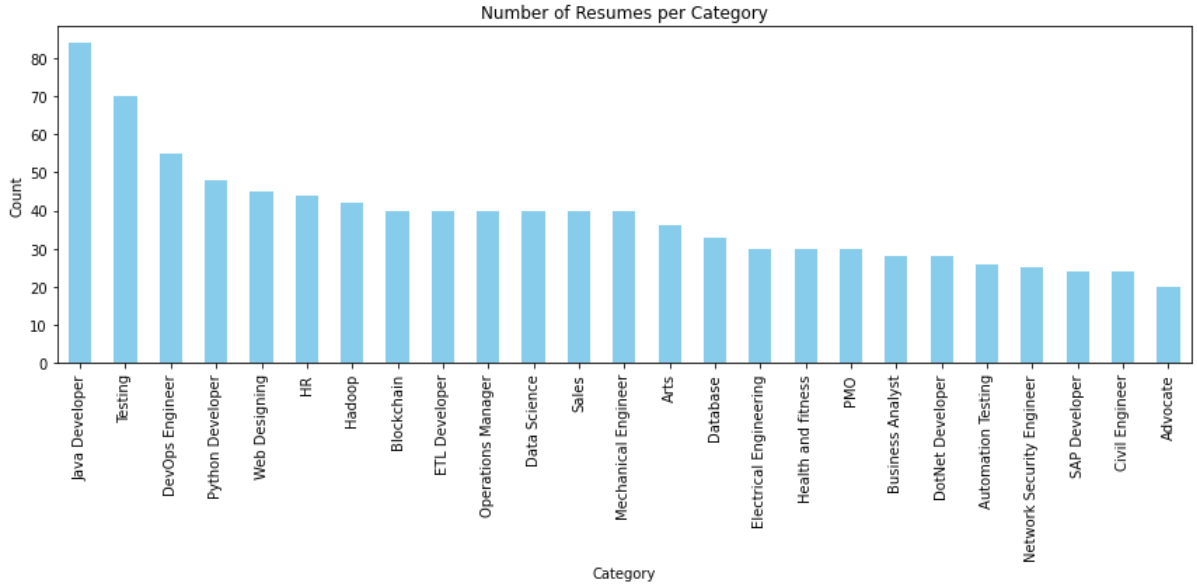
In [5]:
```
print("Total Resumes:", df.shape[0])
print("Total Categories:", df['Category'].nunique())
print("\nCategory Distribution:\n")
print(df['Category'].value_counts())
```

```
Total Resumes: 962
Total Categories: 25

Category Distribution:

Java Developer               84
Testing                      70
DevOps Engineer              55
Python Developer             48
Web Designing                45
HR                           44
Hadoop                       42
Blockchain                   40
ETL Developer                40
Operations Manager           40
Data Science                 40
Sales                        40
Mechanical Engineer          40
Arts                         36
Database                     33
Electrical Engineering       30
Health and fitness           30
PMO                          30
Business Analyst             28
DotNet Developer             28
Automation Testing           26
Network Security Engineer    25
SAP Developer                24
Civil Engineer               24
Advocate                     20
Name: Category, dtype: int64
```

In [6]:
```
plt.figure(figsize=(12,6))
df['Category'].value_counts().plot(kind='bar', color='skyblue')
plt.title("Number of Resumes per Category")
plt.xlabel("Category")
plt.ylabel("Count")
plt.xticks(rotation=90)
plt.tight_layout()
plt.show()
```



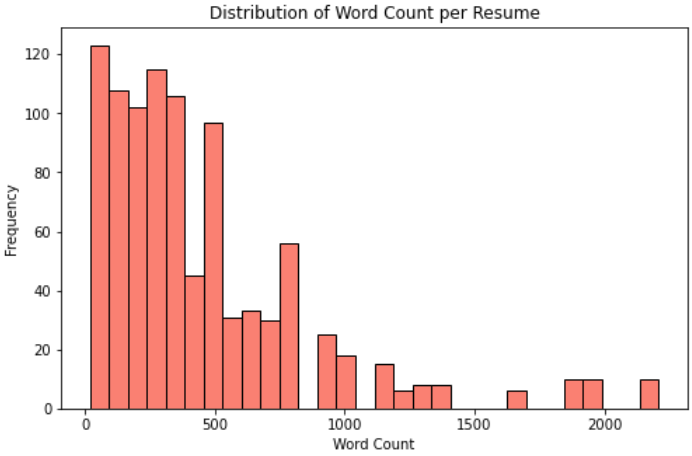
In [7]:
```
df['word_count'] = df['Resume'].apply(lambda x: len(str(x).split()))
df['char_count'] = df['Resume'].apply(lambda x: len(str(x)))

print("Average word count per resume:", df['word_count'].mean())
print("Average character count per resume:", df['char_count'].mean())

plt.figure(figsize=(8,5))
```

```
plt.hist(df['word_count'], bins=30, color='salmon', edgecolor='black')
plt.title("Distribution of Word Count per Resume")
plt.xlabel("Word Count")
plt.ylabel("Frequency")
plt.show()
```

```
Average word count per resume: 450.49792099792097
Average character count per resume: 3160.364864864865
```



In [8]:

```
stop_words = set(stopwords.words('english'))

def preprocess(text):
    text = text.lower()
    text = re.sub(r'[^a-z\s]', ' ', text)
    text = re.sub(r'\s+', ' ', text)
    tokens = word_tokenize(text)
    tokens = [word for word in tokens if word not in stop_words and len(word) > 2]
    return " ".join(tokens)

df['clean_resume'] = df['Resume'].apply(preprocess)
df[['Resume', 'clean_resume']].head()
```

Out[8]:

	Resume	clean_resume
0	Skills * Programming Languages: Python (pandas...	skills programming languages python pandas num...
1	Education Details \r\nMay 2013 to May 2017 B.E...	education details may may uit rgpv data scient...
2	Areas of Interest Deep Learning, Control Syste...	areas interest deep learning control system de...
3	Skills â¯¢ R â¯¢ Python â¯¢ SAP HANA â¯¢ Table...	skills python sap hana tableau sap hana sql sa...
4	Education Details \r\n MCA YMCAUST, Faridab...	education details mca ymcaust faridabad haryan...

In [9]:

```
all_words = " ".join(df['clean_resume']).split()
word_freq = Counter(all_words)

print("Top 20 most frequent words in the entire dataset:\n")
for word, freq in word_freq.most_common(20):
    print(f"{word}: {freq}")
```

```
Top 20 most frequent words in the entire dataset:

exprience: 3829
company: 3578
project: 3498
months: 3288
description: 3122
details: 3096
data: 2156
management: 1999
team: 1950
maharashtra: 1449
system: 1425
testing: 1349
year: 1344
database: 1280
development: 1203
business: 1196
ltd: 1177
test: 1174
less: 1145
using: 1124
```

In [10]:

```
all_text = " ".join(df['clean_resume'])

wordcloud = WordCloud(
    width=1000,
    height=600,
    background_color='white',
    max_words=150,
    colormap='viridis'
```

```
).generate(all_text)

plt.figure(figsize=(12,7))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title("Word Cloud - All Resumes", fontsize=16)
plt.show()
```



In [11]:

```
category_name = "Data Science"

category_text = " ".join(df[df['Category'] == category_name]['clean_resume'])

wordcloud_cat = WordCloud(
    width=1000,
    height=600,
    background_color='white',
    max_words=150,
    colormap='plasma'
).generate(category_text)

plt.figure(figsize=(12,7))
plt.imshow(wordcloud_cat, interpolation='bilinear')
plt.axis('off')
plt.title(f"Word Cloud - {category_name} Resumes", fontsize=16)
plt.show()
```



In [12]:

```
categories_to_compare = ["Data Science", "Java Developer", "HR", "Testing"]

fig, axes = plt.subplots(2, 2, figsize=(16, 10))
axes = axes.flatten()

for i, cat in enumerate(categories_to_compare):
    text = " ".join(df[df['Category'] == cat]['clean_resume'])
    wc = WordCloud(width=600, height=400, background_color='white', max_words=100).generate(text)
    axes[i].imshow(wc, interpolation='bilinear')
    axes[i].set_title(cat, fontsize=14)
    axes[i].axis('off')
```

```
plt.tight_layout()
plt.show()
```



Conclusion